INCH-POUND
MIL-DTL-915/7G
22 August 2002
SUPERSEDING
MIL-C-915/7F
29 September 1989

DETAIL SPECIFICATION SHEET

CABLE, POWER ELECTRICAL, FOR OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE DSWS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-915 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First	-	Copper conductor, AWG size 14, class B stranding, tin coated.
Second	-	Synthetic rubber insulation. Nominal wall thickness 0.073 inch. Standard identification code applied by method 3. An additional covering of clear polyamide may be employed at the manufacturer's option.
Third	-	Two conductors shall be cabled together with a lay not greater than 5 inches. Fillers of synthetic rubber-like compound shall be used to attain a firm, well rounded cross section. Fillers may be laid in the valleys during the cabling operation or may be extruded into the valleys separately. Fillers shall be readily separable from the insulation without damage thereto and without destroying the conductor identification.
Fourth	-	Binder tape over the conductor assembly, at manufacturer's option.
Fifth	-	Shielding braid of AWG no. 34, tin coated copper, with an angle of 30 to 40 degrees and minimum coverage of 88 percent.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Sixth - An overall jacket (black), nominal wall thickness 0.060 inch. if needed to meet electrical tests, the jacket may comprise - inner layer of rubber insulation, such as synthetic rubber and - outer layer of rubber insulation, such as synthetic rubber and - outer layer of polychloroprene. The jacket shall be readily separable from the underlying shield, leaving the shield adequately clean for soldering. Cable surface marking required.

TABLE I. Details.

	Overall diameter	
Type and size	Minimum (inch)	Maximum (inch)
DSWS-4	0.770	0.800

EXAMINATION AND TESTS:

Requirements

Basic electrical:

Conductor resistance - ohms/1000 feet at 25°C, maximum 2.78

Voltage withstand - volts, root mean square, minimum

- Conductor to conductor (5 minutes) 10,000
- Conductor to shield (5 minutes) 10,000
- Shield to water (5 minutes) 500

Voltage withstand - volts, dc, minimum

- Conductor to conductor (15 minutes) 25,000
- Conductor to shield (15 minutes) 25,000
- Shield to water (15 minutes) 1,500

Insulation resistance – megohms-1000 feet, minimum

- Conductor to conductor 200
- Conductor to shield 200
- Shield to water 100

Group A:

- Visual and dimensional ..... No failure
- Hydrostatic (open end) - 1000 lbf/in^2, in 2 hours, in^3, maximum 0
- Mutual capacitance - picofarad per feet (pF/ft) at 1 kilohertz (kHz) ..... 20 ± 10

Group B:

Physicals (unaged)

- Insulation (conductor)
  - Tensile strength - lbf/in^2, minimum 600
  - Elongation - percent, minimum 350
- Jacket (cable)

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Tensile strength - lbf/in^2, minimum 1,800
Elongation - percent, minimum 300
Set - inch, maximum 3/8
Drip - at 95 ± 1°C ..... Zero

Group C:
Physical (aged)
Insulation (conductor)
Air oven test
Tensile strength - percent of unaged, minimum..... 70
Elongation - percent of unaged, minimum..... 75
Jacket (cable)
Air pressure heat
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum..... 50
Hot oil immersion
Tensile strength - percent of unaged, minimum..... 60
Elongation - percent of unaged, minimum..... 60
Permanence of printing (jacket) cycles, minimum..... 250
Cable filler removability ..... No failure
Shield - conformance to material, coverage and construction ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

Insulation resistance per 1000 feet at room temperature from shield to water, 50 megohms, minimum after 90 days continuous submergence in water; values recorded at 1-week intervals.

Flying endurance - 1000 cycles of bending a specimen, 4-inch maximum diameter mandrel at minus 40 ± 2°C. Upon successful completion of this test, specimen shall meet the voltage withstand test and the hydrostatic (open end) test.

UNIT ORDERING LENGTH:
500 feet (nominal).

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PART OR IDENTIFYING NUMBER (PIN): The PIN consists of the letter M, the general specification number along with the specification sheet, and a dash number representing type and size. Refer to MIL-DTL-915 for an example.

M	**915**	**/7-**	**X-X**
Prefix for Military Specification	Specification Number	Specification Sheet Number	Type and Size

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy - SH
(Project 6145-2317-003)

Review activities:
Army – AR, CR
Navy - CG, EC
DLA - CC

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